Photoplethysmography Project

Final Report for ENGE 250
Electrical Circuit Analysis
George Fox University
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PPG Project
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I. PROJECT OVERVIEW

The purpose of this project was to construct a simple photoplethysmography circuit capable of detecting changes in blood flow at the fingertip. A PPG operates by passing light through tissue and measuring changes in light intensity caused by variations in blood volume. In this specific circuit, an LED and an LDR were used together in the human-interface portion of the circuit. They were placed into a finger clamp that created an isolated system that blocked outside light, so the LDR only detected changes in light passing through the fingertip. Since the blood creates more darkness, the resistance in the LDR changes, allowing the voltage to increase. This increase is the raw signal that is passed into the circuit. This project was intended to demonstrate the basic principles used in medical instrumentation while also reinforcing foundational circuit analysis concepts such as amplification, filtering, and signal interpretation.

Since the raw signal was so small, amplification was required for the heartbeat waveform to be clearly observed on the oscilloscope. The circuit had a virtual ground stage, two inverting gain stages, and a low pass filter. The virtual ground stage created a midpoint bias so the op-amps could amplify the heartbeat signal while operating from only a single 9 V supply. Gain Stage 1 and Gain Stage 2 increased the amplitude of the input and were used to drive the graphical LED output. The low-pass filter was included to reduce high frequency noise. All these parts together turned a tiny fingertip sensor signal into a viewable waveform.

In addition to the signal filtering and amplification part of the circuit, there was also the LED graphical display. This stage provided a visual representation of the signal on the circuit board. It used a resistor ladder and comparators to activate the LEDs at different thresholds. This caused the LEDs to turn on in a varying, pulse-like fashion, which looked like a heartbeat pulsing up and down. Since the signal could have varying gain, potentiometers were attached to the comparator network to set the LED trigger thresholds. This part of the project showed how a varying signal could be converted into a discrete LED display.

The finger clamp system was created using a 3D-printed finger clamp shell that had holes to insert the LDR and LED. Appropriate-length wires were then soldered to the components. A spring was then used to provide clamping force onto the finger. The reason for this was that if the user held the finger clamp too tightly on their finger, it could disrupt the signal. This device was simple, yet it proved to be powerful in delivering a human heartbeat signal.

The main purpose of this project was to ultimately show how powerful simple components can be in circuits. The circuit only used about four major elements: LDR, op-amps, comparators, and a filtering capacitor. Yet it was able to capture the waveform of a biological human’s heartbeat using only a small fingertip. That is wildly impressive and incredibly useful. Overall, this project was a great introduction to the immense power of electrical engineering.

II. CIRCUIT

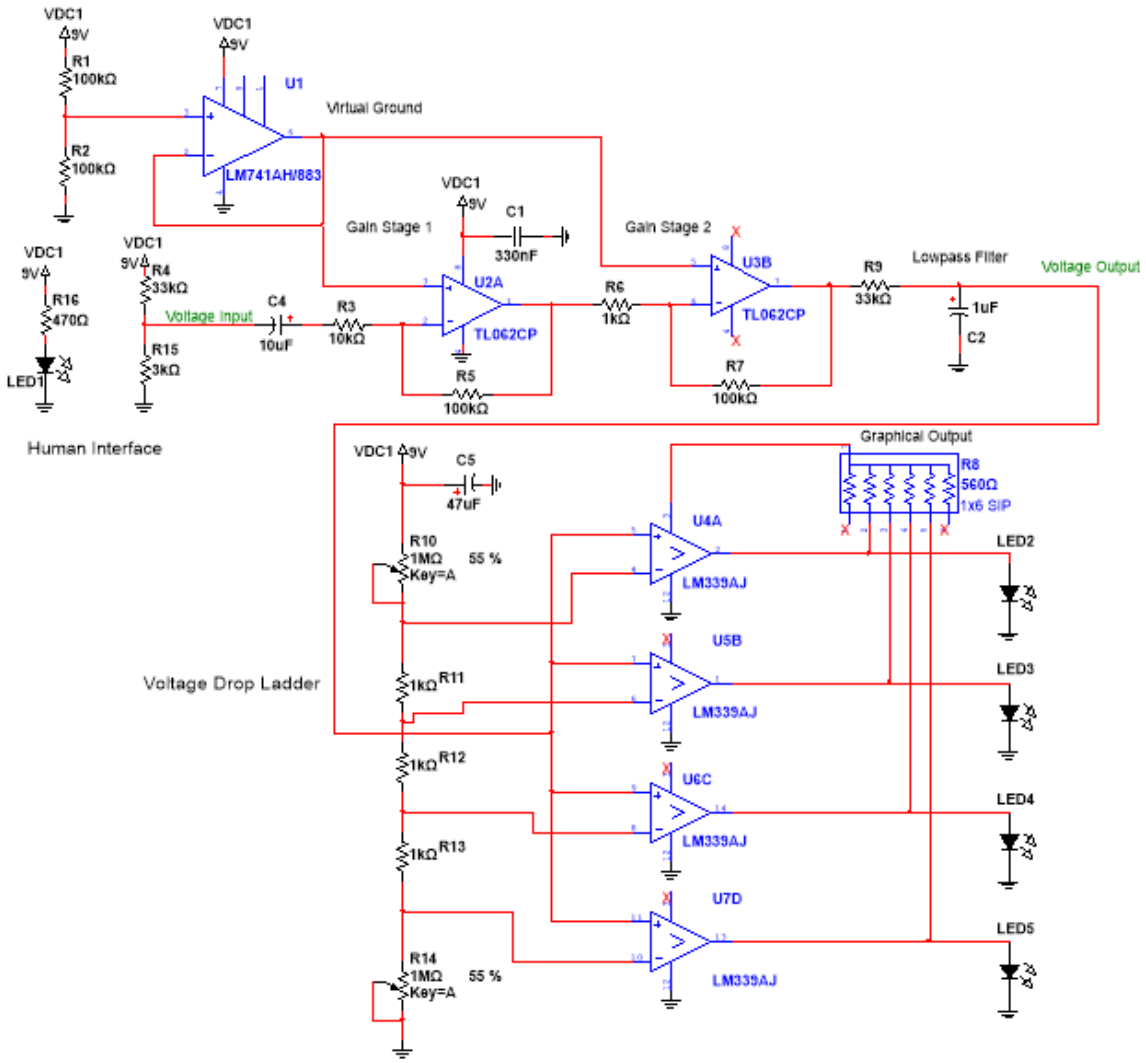


Figure 1. *PPG Circuit*

III. CIRCUIT OPERATION

Stage one of the circuit starts with a voltage divider that delivers half of the source voltage to the LM741. The op-amp acts as a buffer and is intended to make the virtual ground usable. If the resistor divider were by itself, the rest of the circuit would load it down and the 4.5 V reference point would move around. The buffer allows this reference point to remain stable for the rest of the circuit. The purpose of the virtual ground is to allow the amplifiers to swing up and down without needing two 9 V supplies, thus simplifying the circuit's cost and size.

Stage two of the circuit involves amplification of the input signal. The LDR and R4 create a voltage divider with the 9 V source. Capacitor C4 plays a crucial role in filtering out the DC part of the signal. It allows only the changing heartbeat signal to pass into the amplifier. The signal then goes through two gain stages. Because the heartbeat signal is so small, trying to get all the needed gain from one stage is difficult. Having two gain stages makes the circuit easier to test and troubleshoot. The total theoretical gain is 1000.

Stage three of the circuit is the low-pass filter. C2 is responsible for removing high-frequency noise and preserving the slower pulse waveform.

Stage four of the circuit is the graphical output and voltage-drop ladder. The voltage is reduced to different thresholds through the series resistors and potentiometers. These signals are then compared to the varying heartbeat pulse, which turns on the LEDs if the heartbeat signal is greater than the individual thresholds. The 1x6 560Ω SIP reduces the current to the LED's so they don't burn out.

The only modification made to the original circuit was the usage of a 1x6 560Ω SIP. This modification was made due to insufficient 1x4 560Ω SIP inventory.

IV. RESULTS

Table 1. Theoretical and experimental voltage gains of the amplifier stages.

Stage	Theoretical (V/V)	Experimental (V/V)	Percent Difference (%)
Gain Stage 1	10.0	11.1	**11.0%**
Gain Stage 2	100.0	57.2	**42.8%**
Overall Gain	1000.0	635	**36.5%**

Table 2. Theoretical and experimental cutoff frequency of the low-pass filter.

	Theoretical (Hz)	Experimental (Hz)	Percent Difference (%)
Cutoff Frequency	4.82	8.00	**66.0%**

Calculations used for Experimental Values

$$\boldsymbol{A_{v,\text{Stage 1}} = \frac{V_{out}}{V_{in}} = \frac{6.72}{0.600} = 11.2}$$

$$\boldsymbol{A_{v,\text{Stage 2}} = \frac{V_{out}}{V_{in}} = \frac{4.30}{0.0752} = 57.2}$$

Calculations used for Theoretical Values

$$\boldsymbol{A_{v,\text{Stage 1}} = -\frac{R_f}{R_{in}} = \frac{100 \times 10^3}{10 \times 10^3} = -10}$$

$$\boldsymbol{A_{v,\text{Stage 2}} = -\frac{R_f}{R_{in}} = \frac{100 \times 10^3}{1 \times 10^3} = -100}$$

$$\boldsymbol{A_{v,\text{overall}} = A_{v,\text{Stage 1}} A_{v,\text{Stage 2}} = (-10)(-100) = 1000}$$

R = 33kΩ
C = 1μF

$$\boldsymbol{f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(33 \times 10^3)(1 \times 10^{-6})} = 4.82 \text{ Hz}}$$

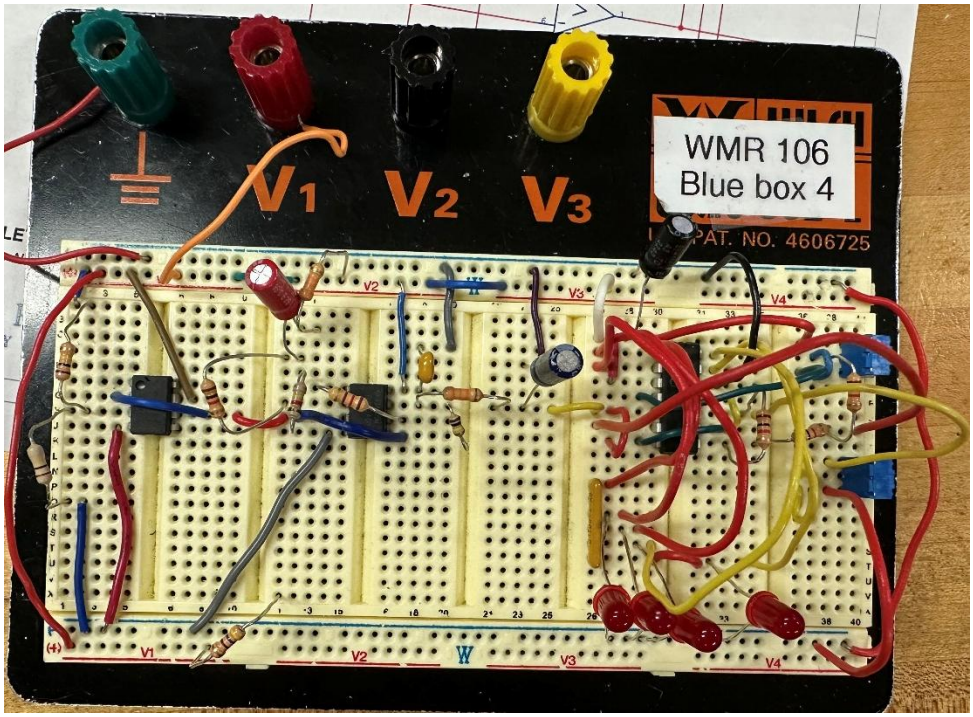


Figure 2. Finished Breadboarded Circuit



Figure 3. Finger Clamp

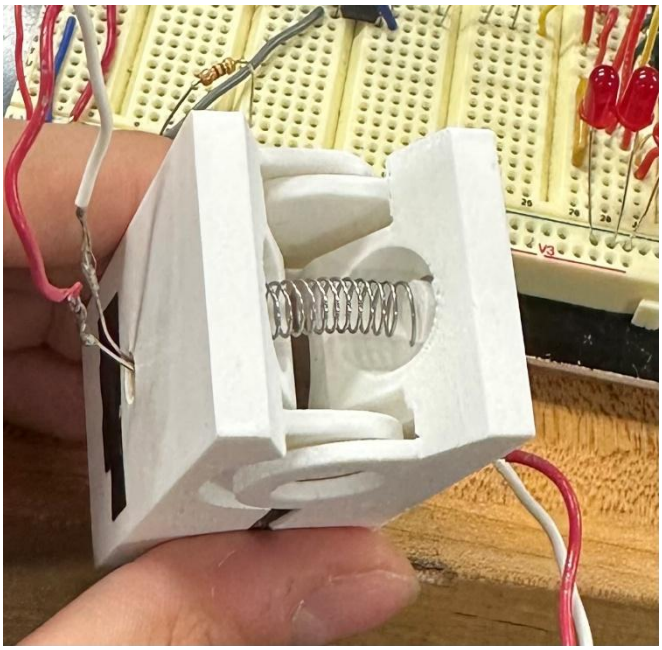


Figure 4. Finger Clamp and Spring

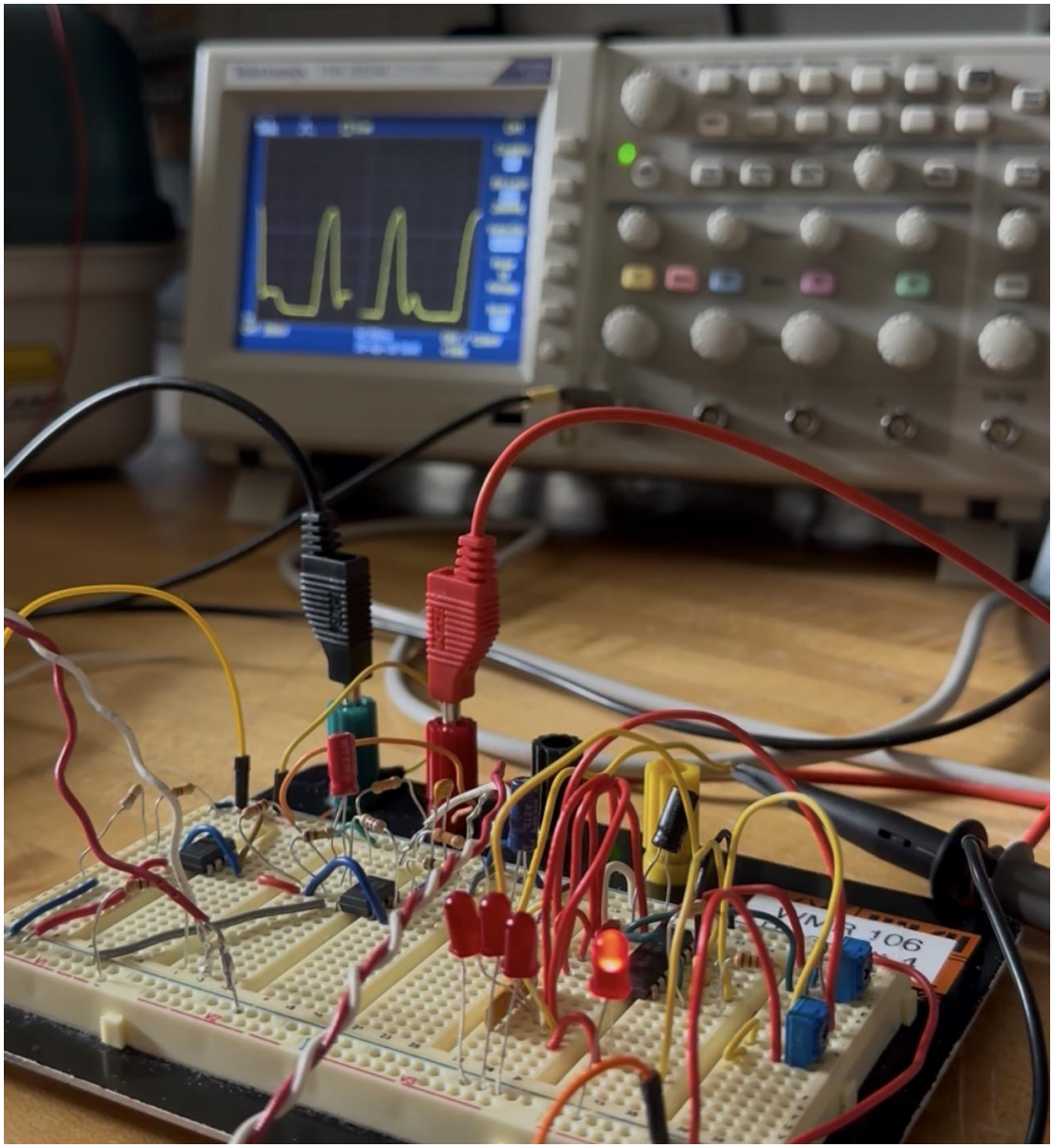


Figure 5. Breadboard implementation of the PPG circuit, with the measured heartbeat waveform displayed on the oscilloscope and the LED graphical display illuminated.

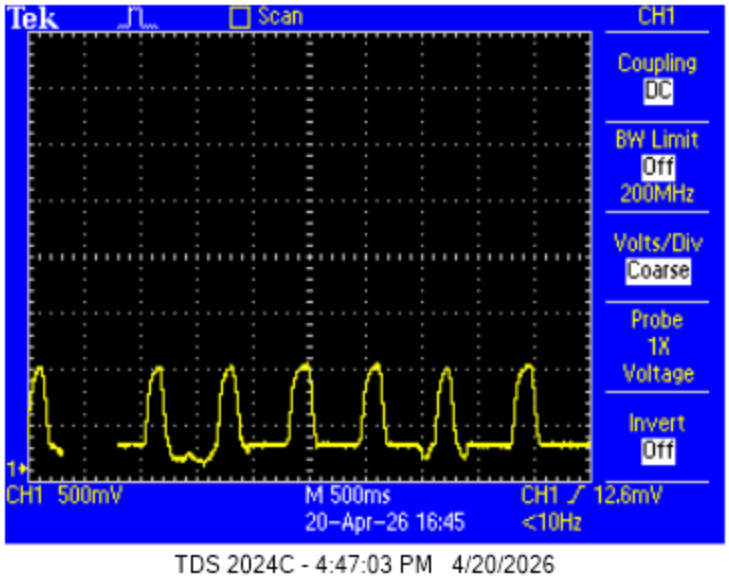


Figure 6. Oscilloscope capture of the PPG output showing a steady repetitive heartbeat waveform at 500 ms/div and 500 mV/div.

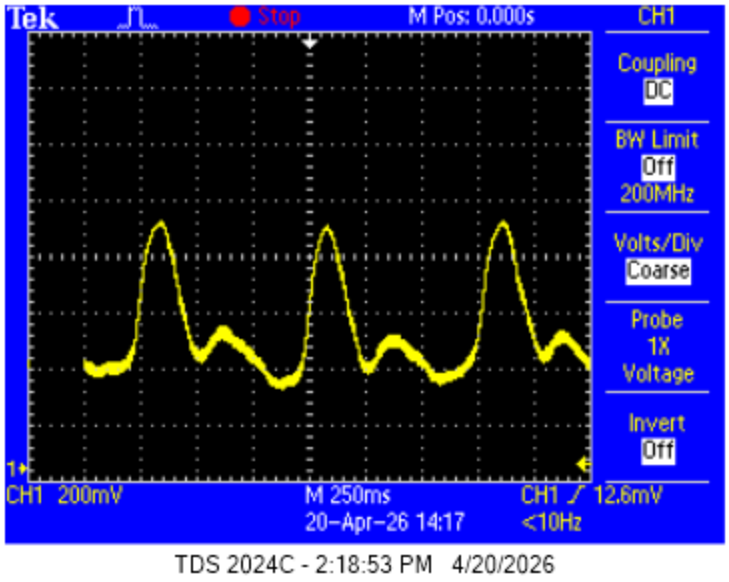


Figure 7. Oscilloscope capture of the PPG output showing a more detailed view of the heartbeat waveform at 250 ms/div and 200 mV/div.

V. DISCUSSION

The project was successful in demonstrating that a simple PPG circuit could detect and clearly show a heartbeat waveform from the fingertip. The circuit that was constructed matched theory in most areas. The main discrepancies were in the gain stages. The theoretical overall gain of 1000 was not achieved; when tested with a 1.2 Hz square wave, the circuit produced an overall gain of approximately 635. This was not ideal, but it still showed that the circuit was roughly working. However, when the heartbeat signal was used as V_{in}, the output was only about 0.8 V peak. With the goal of increasing the peak output, resistor ratios were changed to increase the gain. Changing the resistor ratios barely increased the gain, and reaching 1 V peak was the highest result achieved. Even trying a different IC did not significantly change the results.

Testing the finger clamp in darkness produced the highest voltage. In darkness, the oscilloscope read about 1.5–2.0 V peak. This suggests that a significant amount of ambient light was getting into the finger clamp and disrupting the results. The graphical display was also very difficult to get working when the voltage pulses were so small and sharp. In darkness, and with careful potentiometer adjustment, the graphical display worked much better. The conclusion reached was that ambient light leakage was the main problem rather than the op-amp not providing enough gain. The circuit was simply not receiving a clean signal.

Another observation was finger placement. The finger needed to be in a very precise position, and any slight movement could disrupt the signal. Even moving the opposite arm affected the signal, and standing also caused noticeable interference.

Overall, the circuit did work and clearly showed a heartbeat waveform. The graphical display was finicky, but it worked with proper adjustment.

VI. Conclusion

The design could be improved upon with a better finger clamp that blocks more ambient light and locks the finger in better. The design is also difficult to implement onto a breadboard because it introduces opportunities for noise and poor contacts. Even the lengths of all the wires play a role with such a small signal. A PCB design would be much more appropriate here. Not that a PCB design would automatically fix the issues directly, but it would improve consistency and connections. Which would further troubleshooting success because it makes finding the main issue easier.